



MUHAMMAD RAIHAN ALIFIANDRA

Instrumentation, Control Systems & Electronics Engineer

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Pekanbaru, Riau, Indonesia

A 5th-semester Electronics Engineering Technology student at Politeknik Caltex Riau, specializing in instrumentation and control systems. I design closed-loop PID control systems, perform sensor calibration, and build signal conditioning circuits on Arduino and ESP32, with LabVIEW and Python for data acquisition and monitoring. I also have foundational in PLC programming (SoMachine Basic) and SCADA/HMI development (Wonderware InTouch). As a research assistant and Research and Technology Division Coordinator, I work collaboratively in research teams and communicate technical work clearly.

My technical foundation includes data acquisition with Raspberry Pi, Arduino, and ESP32, LabVIEW and Python for monitoring, and Multisim for circuit validation.

Education

Politeknik Caltex Riau - Pekanbaru, Riau, Indonesia Sep 2024 - Aug 2028 (Expected)
Bachelor of Electronics Engineering, 3.79/4.00

- Relevant Courses: Instrumentation, PLC, Control Systems, and Electrical Circuit Analysis
- Research Assistant of IoMT & AI-Based Arrhythmia Monitoring System (data collection assistant)
- PKM – Digital Prayer Schedule Display System on Raspberry Pi, Masjid Baitul Maqdis (Electronics Technician)
- Robotics Training Instructor for P5 Program at SMAN 17 Pekanbaru

SMAN 2 BALIKPAPAN - Balikpapan, Kalimantan Timur, Indonesia Sep 2020 - May 2023
High School Diploma in Science, 90.25/100.00

- Documentation Staff, Mathematics Enthusiast Community
- Member, Islamic Student Organization (Rohis)

Projects

Multi-Stage Signal Conditioning Circuit

- Designed a 4-stage analog signal conditioning circuit (non-inverting amplifier with $11\times$ gain, Sallen-Key low-pass filter with $f_c \approx 15.9$ Hz, voltage-divider attenuator, voltage-follower buffer) to convert the LM35 output into an ADC-ready signal for an Arduino Nano.
- Used a TL431 as a 2.5 V external AREF, improving ADC resolution from ~ 0.5 °C/step to ~ 0.109 °C/step, and added reverse polarity protection with a 1N4007 diode.
- Validated the system against a reference thermometer at 5 points (20–100 °C): error of 0.34–1.57 °C (1.54%–1.68%), within the calculated theoretical range.

Aero Pendulum

- Designed a closed-loop PID control system on Arduino Uno and MATLAB Simulink to hold an underactuated aero pendulum at a target angle using propeller thrust, with a potentiometer for angle feedback
- Calibrated the angle sensor with a two-point ADC-to-degree conversion, then tuned the PID with Ziegler-Nichols and manual fine-tuning ($K_p = 0.95$, $K_i = 1.3$, $K_d = 0.22$)
- Achieved overshoot of 3–8%, settling time of 4–4.5 s, and steady-state error of 0.5°–2° across 45°, 60°, and 90° setpoints
- Verified disturbance rejection: the arm returned to the setpoint within ± 1.2 s after a manual push, without sustained oscillation

Real-Time Lab Environment Monitoring

- Developed an ESP32-based monitoring system for temperature, humidity, and pressure (AHT20, BMP280 via I2C) with threshold-based alarms
- Implemented a LabVIEW interface with real-time indicators and charts via serial communication, complemented by an MQTT-based web dashboard
- Validated sensor accuracy against reference instruments with average errors of 0.3°C, 1.58 %RH, and 0.56 hPa, within datasheet tolerance

Skills

Electrical Circuits

Circuit Analysis (DC/AC), Analog Circuit Design, Circuit Simulation

Measurement & Testing

Multimeter, Oscilloscope, Function Generator, Power Supply, Circuit Troubleshooting

Instrumentation

Sensor Calibration, Signal Conditioning, Data Acquisition

Control System

PID Control, Closed-Loop Feedback, Control System Design

Embedded System

Arduino, ESP32, Raspberry Pi, MQTT, C/C++, Python, MySQL

Tools & Software

LabVIEW, Multisim, MATLAB, SoMachine, Wonderware Intouch Arduino IDE, Visual Studio Code, Github

Soft Skills

Leadership, Communication, Critical & Analytical Thinking, Teamwork

Organizational Experience

Intelligent Control by Human Politeknik Caltex Riau

Aug 2024 - Present

Research and Technology Division Member

A student organization focused on robotics, embedded systems, and intelligent control innovation.

- Conducted research on emerging technologies for workshop and training program development
- Selected components and designed wiring schematics/technical diagrams for workshop references
- Delivered technical materials and facilitated knowledge-sharing sessions for participants

Electronics Engineering Student Association, Politeknik Caltex Riau

Sep 2025 - Present

Research and Technology Division Coordinator

The official student association representing the Electronics Engineering Study Program at PCR.

- Led and coordinated division members in research activities and technical responsibilities
- Initiated and led theoretical learning sessions on foundational and applied electronics concepts
- Oversaw component selection and technical documentation for hands-on workshop implementation